IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2025

CIRCUITS AND SYSTEMS

Tuesday, 13 May 2025, 2:30 PM

Time allowed: 2 hours

SOLUTIONS

There are THREE questions on the paper.

Answer ALL questions.

SOLUTIONS

1. (a) This question tests student's ability to use an op-amp and demonstrate their understand of its limitations.

(i) (Bookwork, but from first principle). C2 blocks dc from Vs. Therefore, the gain of the dc offset of Vs is 0.

R3 and R4 forms a voltage divider and must be equal to ac signal of Vs.

$$Vout * \frac{R3}{R4 + R3} = Vs$$

Hence the AC gain is (1+R4/R3) = 2.33.

Also, student may derive the equation relating Vout to Vs:

$$Vout = \frac{R4 + R3}{R4} * \left(Vs + 3.3 * \frac{R2}{R1 + R2}\right) = 2.33 * Vs + 2.56$$

[4]

(ii) V+ dc voltage is determined by R1 and R2 and it is 1.1V. The gain of the amplifier is 2.33 for both dc and ac at V+. Therefore, Vout an ac signal with pk-pk voltage of 2.33 x 1V = 2.33V, with a dc offset of 2.33 x 1.1V = 2.56V. Therefore Vout(max) = 3.73V and Vout(min) = 1.4V. Unfortunately, the power supply is at 3.3V.

Therefore, Vout is clipped for any voltage above 3.3V.



[3]

(iii) R1 and R2 gives a total resistance of R1 || R2 = 6.67k, with C2, the corner frequency is 24Hz.

Therefore, 1Hz sine wave does not pass through C2 unimpeded. Since the GBP is 1MHz, it also cannot provide a gain of 2.33 at this signal frequency. Therefore, the gain of the amplifier is lower than the predicted 2.33.

[3]

SOLUTIONS

(b) This question tests student's understanding pseudo random binary sequence generator circuit and how to implement such component in SystermVerilog..

(i) The signature for zero input is:

Cycle	Q[5:1]
1	00001
2	00010
3	00101
4	01010
5	10101
6	01011

[3]

(ii) When merging input sequence, the signature is:

Cycle	in	Q[5:1]
1	1	00001
2	0	00011
3	1	00100
4	1	01001
5	0	10011
6	0	01010

[3]

(iii)

```
module signature (
    input  logic        clk,    // clock
    input  logic        init,   // reset signature to initial state
    input  logic        in,     // input signal
    output logic [5:1]  Q,      // output signature
);

    // state registers
    always_ff @(posedge clk)
        if (init)    Q <= 5'b00001;
        else    Q <= {Q[4:1],Q[5]^Q[2]^in};

endmodule
```

[4]

SOLUTIONS

(c) This question examines student's understanding of propagation delay in digital circuit and how to calculate the maximum clock frequency in a sequential circuit..

(i) P0 to C2A = 2 + 2 + 2 = 6ns
C-1 to C2A = 2 + 2 + 2 = 6ns

$P0 \rightarrow C2B$ = max(2 + 2 + 2 + 2 , 2 +1+ 3) = 8ns. WORSE CASE DELAY.

$C\text{-}1 \rightarrow$ = C2B =2ns

The following explanation is not expected, but it is included here for educational purpose in future years.

Note that although there is an apparent path $C\text{-}1 \rightarrow C0 \rightarrow C1 \rightarrow C2A \rightarrow C2B = 8$.

This in fact never actually arises - it is never true that a change in C-1 will cause this chain of cause and effect because it would require all the full-adders to be propagating the carry (to enable $C\text{-}1 \rightarrow C0 \rightarrow C1 \rightarrow C2A$ *).*

Also, SEL=0 (to enable the final step C2A → C2B). The entire point of the circuit (known as carry-skip adder) is that we prevent this from happening. Normally when analysing a circuit for the worst-case propagation delay, we assume that any path that is present in the circuit will be enabled for some set of input conditions; the carry-skip circuit is one of the rare cases when this assumption is untrue.

[6]

(ii) The worst-case path of the adder is $P0 \rightarrow C2B$ = 8ns.

P0 is the output of the pipeline register from the stage before. Therefore, it arrives 1.5ns after the clock edge. The setup time is 2ns. Therefore, the clock period Tc must be:

$$Tc \geq 8 + 2 + 1.5 \text{ ns}.$$

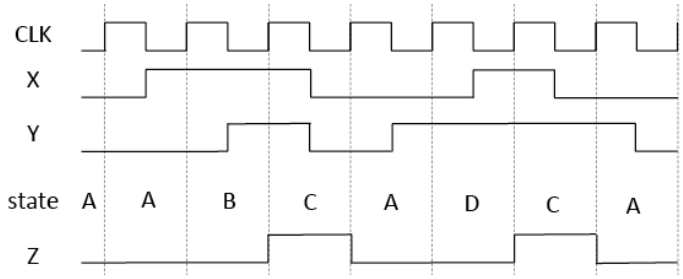
Hence maximum clock frequency is 1000/(11.5) MHz = 87MHz. Hold time does affect the operation of the circuit.

[4]

SOLUTIONS

(d) This tests students' ability to analyse a state diagram and from this specify its implementation in SystemVerilog.

(i)



[4]

(ii) A sample SystemVerilog design:

```
module FSM (
    input  logic    RST,    // reset input
    input  logic    CLK,    // clock
    input  logic    X, Y    // input signals
    output logic    Z       // output signal
);

    // define states
    typedef enum {S_A, S_B, S_C, S_D} my_state;

    my_state   current_state, next_state;

    // state registers
    always_ff @(posedge clk)
        if (rst)    current_state <= S_A;
        else        current_state <= next_state;

    // next state logic
    always_comb
        case (current_state)
            S_A:    if (X==1'b1) && (Y==1'b0) next_state = S_B;
                    else if (X==1'b0) && (Y==1'b1) next_state = S_D;
                    else next_state = S_A;
            S_B:    if (X==1'b0) && (Y==1'b0) next_state = S_A;
                    else if (X==1'b1) && (Y==1'b1) next_state = S_C;
                    else next_state = S_B;
            S_C:  next_state = S_A;
            S_D:    if (X==1'b0) && (Y==1'b0) next_state = S_A;
                    else if (X==1'b1) && (Y==1'b1) next_state = S_C;
                    else next_state = S_D;
        default: next_state = current_state;
        endcase

    // output logic
    always_comb
        if (current_state == S_C) Z = 1'b1;
        else Z = 1'b0;

endmodule
```

[6]

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(e) This questions tests student's understand of flash ADC, voltage range and simple encoder to translate the thermometer code to binary code.

(i) Assumptions: all resistor values R are identical and the comparators have zero offset error.

ABCD	Vin
0000	Vin ≤ 0
0001	0 ≤ Vin < 1
0011	1 ≤ Vin < 2
0111	2 ≤ Vin < 3
1111	Vin > 3

[5]

(ii)

```
module encoder (
    input logic A, B, C, D,     // input signals
    output logic [2:0]  X;      // output signals
)
    always_comb
        case {A,B,C,D}
            4'b0000: X = 3'd0;
            4'b0001: X = 3'd1;
            4'b0011: X = 3'd2;
            4'b0111: X = 3'd3;
            4'b1111: X = 3'd4;
        default
            X = 3'd0;   // should never happen
        endcase
endmodule
```

[5]

SOLUTIONS

2. This question tests student's ability to read datasheet of an op-amp, and to demonstrate their understanding of the implications of the various parameter and thus what to expect from the performance of the op-amp. It further tests student's ability to derive SPICE model parameters from the op amp specification.

(a) The relevant parameters are:

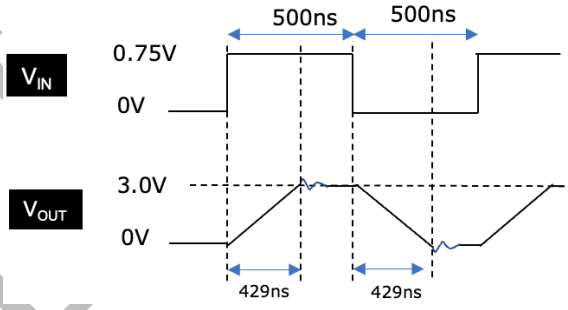
Parameters	Sym	Min	Typ	Max	Units	Conditions
AC Response						
Gain Bandwidth Product	GBWP	—	10.0	—	MHz	
Phase Margin at Unity-Gain	PM	—	65	—	°	G = +1 V/V
Slew Rate	SR	—	7	—	V/µs	

Assumption: we use typical parameter.
This is a non-inverting x4 amplifier. Therefore, ignoring gain-bandwidth product limitations, the output should be a sinewave with dc offset of 2V and amplitude of 1V. Since the supply is 3.3V, and MCP6291 has rail-to-rail capability, the output can swing between 1V to 3V without clipping.

However, the gain bandwidth product is only 10MHz. With a gain of 4, the break frequency of the x4 amplifier is expected to be at 2.5MHz. The gain at this frequency is 3dB (i.e. 0.707) lower than the expected gain of 4.
Therefore, the amplitude is drop to 0.7V. Hence the correct answer would be between 1.3V and 2.7V.

[6]

(b) The slew rate is 7V/us, therefore it takes 429ns to slew 3.0V. Hence the output waveform is as shown here:



[4]

(c) For maximum output voltage range, VOUT should be 3.3V when E_θ is 1mW/cm^2. From the photodiode characteristics, the reverse current is at about 70µA. Therefore, R should be 3.3V/70µA = 47kΩ.

[3]

(d) (i) From datasheet $Z_{DIFF} = 10^{13}\Omega \parallel 3pF$, R1 = 1TΩ, C1 = 3pF, V1 = V_{OS} = 3mV.

[2]

(ii) GBP = 10M. Open-loop gain A_{OL}=110dB or 400,000.
Dominant pole $f_p = \frac{GBP}{A_{OL}} = \frac{10^7}{400000} = 25Hz = 1/2\pi R_2 C_2$.
Assume $C_2 = 1nF$, R2 = 6.37MΩ.
$gm \times R_2 = A_{OL}$, hence $gm = \frac{400{,}000}{6.37} \times 10^{-6} = 0.0628$.

[6]

SOLUTIONS

(iii) Slew rate = max(dV_{C2}/dt) = $Imax_{G1}/C2$. $Imax_{G1}$ = 7V/us * 1nF = 7mA. Therefore, the SPICE statement for G1 is:

G1 0 int_gain value={limit(0.0638*V(offset, in-), 7m, -7m)}

[3]

(iv) D1 and D2 are to limit the output voltage to that of V+ and V-.

[1]

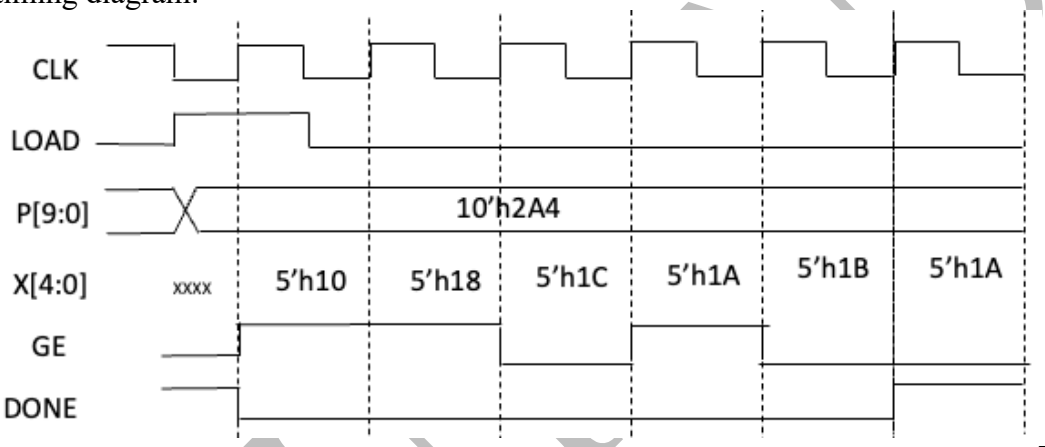
SOLUTIONS

3. This question tests student's understanding the principle of successive approximation algorithm and how to apply it. It also tests student's ability to design a reasonably complicated FSM in SystemVerilog.

(a) Although not required by the question, it could be easier to draw up the following table:

X[4:0]	X^2	GE	RESULT
5'h10	10'h100	TRUE	X[4] = 1
5'h18	10'h240	TRUE	X[3] = 1
5'h1C	10'h310	FALSE	X[2] = 0
5'h1A	10'h2A4	TRUE	X[1] = 1
5'h1B	10'2C9	FALSE	X[0] = 0

The timing diagram:



[10]

(b) The principle of SAR is really bookwork. However, the application here is rather special. Instead of the DAC in a SAR based ADC, we use the multiplier circuit to perform the squaring function. The rest of the SAR algorithm is exactly the same as in the successive approximation DAC.

The assumptions are: 1) LOAD is asserted just before CLK going high, 2) P[9:0] is stable throughout the computation.

[5]

(c) The design of the FSM in SystemVerilog will depend on students.

The best solution would be to have idle state, following by 5 states, each for setting each bit of X from MSB to LSB. One can also use X[4:0] and DONE as the state variable.

The poor solution, although works, is to create 33 states, one for IDLE, waiting for LOAD to activate, followed by all possible state for 5 bits output.

[10]

SOLUTIONS